Code No: 123CT **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2017
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) Convert the hexadecimal member 68BE to binary and then from binary to octal number. [2]
b) Write about error detecting and error minimizing codes with examples. [3]
c) Define pair, quad and octet with examples in K – maps. [2]
d) Explain SOP and POS of Boolean functions. [3]
e) Write the logic diagram and truth table of a Full – adder. [2]
f) What is an encoder? Write the truth table of a parity encoder. [3]
g) What are sequential circuits? Give some examples. [2]
h) Explain the working of a 4 – bit ripple counter with help of truth table. [3]
i) List various types of semiconductor memories and write their uses. [2]
j) What are programmable logic arrays? What are their applications? [3]

PART-B

(50 Marks)

2.a) Explain various number systems used in digital logic design with examples and their representation with examples.
b) Simplify the following Boolean functions to a minimum number of literals
(i) $(x+y)(x+y')$ (ii) $xy+x'z+yz$ [5+5]

OR

3.a) Explain about signed binary number and Floating point representation of number system with examples and discuss their advantages.
b) Express the Boolean function $F = A + B'C$ in a sum of Minterms and write its truth table. [5+5]

4.a) Explain the design procedure for 4 – variable K – map and discuss its simplification methods using prime implicants methods.
b) Simplify the Boolean function using K- map method. [5+5]
F (A, B, C,D) = $\sum$ (0,1,2,4,5,6,8,9,12,13,14)

OR

5.a) Implement the following Boolean function with NAND gates after simplification
F (A,B,C) = (1,2,3,4,5,7)
b) Simplify the following Boolean function in sum of products and product of sums
F (A,B,C,D) = $\sum$(0,1,2,5,8,9,10) [5+5]

6.a) Design a combinational circuit hat converts a 4-bit Gray code to a 4bit binary number Implement the circuit with XOR gates.

b) Design a BCD to decimal decoder using the unused combinations of the BCD code as don't care conditions. [5+5]

OR

7.a) Explain the design procedure for constructing BCD adder – subtractor circuit using BCD adder and 9's complementer block diagrams.

b) Construct a 4 to 16 line decoder with five 2 – to -4 – line decoder with enable. [5+5]

8.a) Explain the state equation, state table and state diagrams of sequential circuit with block diagram.

b) Draw the block diagram of a 4 bit synchronous Binary counter using J-K Flip-slops. [5+5]

OR

9.a) Explain the analysis of clocked sequential circuit using T Flip Flops with help of circuit diagram, state diagram and state table.

b) Draw the block diagram of a 4 bit universal shift Register and explain its working with its function table. [5+5]

10.a) Explain about memory decoding and address multiplexing in memories with the help of neat diagrams.

b) Draw the logic diagram of a Programmable Logic Array with 3 inputs, 4 product terms and 2 outputs. [5+5]

OR

11.a) Discuss various types of sequential programmable devices with block diagrams.

b) Tabulate the PLA programming table for the following Boolean functions and minimize the number of product terms. [5+5]

$A(x,y,z) = \sum(1,2,4,6)$
$B(x,y,z) = \sum(0,1,6,7)$
$C(x,y,z) = \sum(2,6)$
$D(x,y,z) = \sum(1,2,3,5,7)$

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